▼ Syntax in Python ▼

Syntax refers to the structure of the language (i.e., what constitutes a correctly-formed program)

Syntax is about the correct arrangement of words and symbols in a code.

example 1: python is case sensitive

```
age = 20
Age = 30
AGE = 40
print(age)
print(Age)
print(AGE)
```

```
20
30
40
```

example 2: python indentation

python uses **indentation** to indicate a block of code, consistent of 4 spaces or a tab

```
if 5 > 2:
    print("Five is greater than two!")
```

```
Five is greater than two!
```

example 3: comments in python

```
# comments are used to explain code, and to make the code more readable
there are two ways to add comments in python

# the first way is to use the hash character # at the beginning of a line
# this is a comment

# the second way is to use triple quotes, either ''' or """
'''
This is a comment
written in
more than just one line
'''
```

example 4: line continuation

python uses the backslash \ to indicate that a statement continues on the next line

```
print("Hello \
World")
```

example 5: multiple statements on a single line

the semicolon ; allows multiple statements on the single line given that neither statement starts a new code block

```
print("Hello"); print("World")
```

example 6: Whitespace within lines

```
a = 1+2;print(a)
a = 1 + 2; print(a)
a = 1 +       2 ; print(a)
```

example 7: parentheses

python allows the use of parentheses to indicate the order of operations

```
print(2 * (3 + 1))
```

▼ Semantics in Python ▼

Semantics of a language involve the meaning of the statements

```
var = 10
print(type(var))

var = 10.5
print(type(var))

var = "Hello"
print(type(var))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
```

```
var1 = 10; var2 = 20
print(var1 == var2)
print(var1 != var2)
print(var1 < var2)
print(var1 > var2)
```

```
False
True
True
False
```

▼✸ Conclusion✸▼

Semantics is about the meaning of the code or logic behind the code, and syntax is about the structure of the code